Analysis of Australian Federal and State Policies Addressing the Decline in 'Critically Endangered' Scalloped Hammerhead Sharks

ENVIRONMENTAL POLICY (ENST90005)

Word Count: **4,204 words**

1. Introduction

Shark conservation has emerged as a critical biodiversity issue as global shark populations rapidly decline, driven by climate-change induced habitat loss and high demands for shark fins in international markets. According to Ferretti et al. (2020, p. 1), around 27 to 73 million sharks are traded for their meat and fins across global markets per year. Furthermore, as per the 2024 Living Planet Index, monitored vertebrae populations have declined by a staggering 73% between 1970 to 2020 (World Wildlife Fund, 2024, p. 7). This indicates a further 4% decline in vertebrae populations over the two years since the 2022 Index was released, which estimated it to be 69% between 1970 to 2018 (World Wildlife Fund, 2022, p. 12). The 2024 Living Planet Report (p. 7) further accounts that the Asia-Pacific (APAC) region, which Australia is a part of, has seen one of the fastest declines in its wildlife populations at 60%, with habitat degradation and overexploitation for the human food system being two of the primary causes. Oceanic shark populations have not been an exception in this scenario with a reported 71% fall in their numbers in the APAC region, mostly due to overfishing (Pacoureau et al., 2021, p. 568). According to Magnuson (2023, p. 1), more than 90% of existing shark catches are 'biologically unsustainable'. This means that the number of sharks caught in these catches far exceeds the numbers required to sustain their populations (Magnuson, 2023, p. 1). Scalloped hammerhead sharks, scientifically known as *Sphyrna lewini*, are critical to Australia's marine ecosystems. They are found globally and across Australia, specifically in the states of New South Wales, Queensland, Western Australia and the Northern Territory – preferring tropical and temperate oceans (CSIRO, 2022, p. 1). As apex predators, they play a vital role in the country's marine food web. In healthy oceans, they maintain the balance of the marine ecosystem by not only keeping prey populations at an equitable distribution but by also eating the sick and injured (Johnson, 2023, p. 4). Furthermore, scalloped hammerhead sharks are essential to keep coral reefs and seagrass beds healthy while also maintain healthy fish stocks (Johnson, 2023, p. 4). However, their populations continue to decline rapidly despite being listed under Appendix II of the Convention of International Trade in Endangered Species of Wild Fauna and Flora (CITES) since 2014, which limits their international trade through monitoring and permits. They are also listed as 'critically endangered' on the International Union for Conservation of Nature's (IUCN) Red List of Threatened Species (Pacoureau et al., 2021, p. 589). Current estimates by the Australian Marine Conservation Society (2023a, p.1) show that their populations have declined up to 80% in Australian waters. Therefore, there is an urgent need for policy change in the way current environmental policies are being implemented in the country – to prevent scalloped hammerhead sharks from going extinct.

This analysis aims to explore the relevant policies implemented, both nationally and internationally, in the context of declining rates of scalloped hammerhead sharks in Australian waters. While the focus of the analysis is primarily on Australian domestic policy, it also examines two important international treaties, CITES and the Convention on the Conservation of Migratory Species of Wild Animals (CMS), as Australia is a party to both agreements (Department of Climate Change, Energy, the Environment and Water, 2024a, p. 1). The analysis conducts a thorough problem identification, analysing the main causes for the issue, i.e., (i) the international shark fin trade, (ii) overexploitation through bycatch in commercial fisheries and poaching, and the (iii) lack of adequate legislative protection of scalloped hammerhead sharks. The next section analyses key environmental policies which impact scalloped hammerhead shark populations in Australia, starting with:

(i) **The Environment Protection and Biodiversity Conservation (EPBC) Act 1999**, which is the country's central environmental legislation that guides the Commonwealth's response to biodiversity protection.

(ii) **The Action Plan for Sharks and Rays 2021** which provides a comprehensive review of the extinction risk of the 328 species of cartilaginous fishes (sharks, rays and chimaeras) in Australia.

(iii) **The Fisheries (Hammerhead Sharks) Amendment Declaration 2022** which introduces specific regulations aimed at protecting hammerhead sharks from illegal, unreported and unregulated (IUU) fishing activity and its subsequent trade in Queensland, and **(iv) Fins Naturally Attached (FNA) Policy.**

2. Problem Identification

This section outlines three of the primary causes which has led to the decline in scalloped hammerhead sharks in Australian marine ecosystems. The combination of high demand for shark fins, ineffective regulatory policy frameworks and ongoing commercial and overfishing practices are identified as the three main reasons which have severely affected their numbers.

2.1. The International Shark Fin Trade

Approximately 1 to 3 million scalloped hammerhead sharks are killed yearly to support the international fin trade (Chapman et al., 2009, p. 224; Magnuson, 2023, p. 1). The fins of scalloped hammerhead sharks are considered to be large and have a ‘high fin-needle’ count, which makes them highly valuable and increases their bargaining power in retail markets (Magnuson, 2023, p. 1). Furthermore, ‘live shark finning’ is the practice of cutting fins from alive sharks and releasing them back into oceans (Australian Marine Conservation Society, 2023, p. 1). Without their fins, sharks are unable to swim properly and will eventually sink to the bottom of the ocean and suffocate (Australian Marine Conservation Society, 2023, p. 1). However, despite ‘live shark fining’ being legislated across Australia as illegal, international trading of shark fins is still permitted in the country. Furthermore, the country plays a key role in the international shark fin trade, ranking as the 17th largest importer of shark fins – importing and exporting 61,634 and 51,019 individual sharks, respectively each year (Murphy, 2024, p. 3). Other than Western Australia, who are still formulating their Fins Naturally Attached (FNA) policy (Western Australian Government, 2023, p. 1), the rest of Australia has successfully implemented FNA policies in their states. However, stricter controls are needed on the import and export of shark products to ensure actual protection of their declining populations.

2.2. Overexploitation / Overfishing

The Marine Stewardship Council (n.d., p. 1) describes ‘overexploitation’ as one of the major reasons behind the loss of ocean biodiversity, globally. According to the World Wildlife Fund

(2024, p. 28), overexploitation or overfishing can further occur in two ways: (i) direct overexploitation for either subsistence or trade through unsustainable fishing or poaching, and (ii) indirect overexploitation when 'non-target species' are accidentally killed, such as bycatch in fisheries. Scalloped hammerhead sharks are among the most at-risk species for both of these types of overexploitation, through overfishing for their fins and bycatch in commercial fishing. Furthermore, research conducted by Ferretti et al. (2020, p. 4) explores how resistance to reduce bycatch in commercial fisheries increases when market-oriented environmental governance mechanisms are in place, such as the rights of fishers to legally sell bycatch shark fins on the grounds of 'preventing resource waste' – further commodifying natural resources. This, in turn, ends up encouraging more shark meat production as it opens up an avenue for the sale of fins which have a high market-value (Cruz et al., 2021, p. 4). Additionally, it also results in higher shark mortality when regulation requires that fins can only be sold when they are attached to the shark, or a certain percentage of the shark's weight must be from the carcass (Cruz et al., 2021, p. 4). Allowing fishers to legally sell bycatch fins, thereby, enables economic profit from an otherwise restricted or heavily regulated practice (Drakopulos, 2020, p. 106). The approach further favours neoliberal market freedoms over strict environmental regulations, and consequently allowing fishers to capitalise on legal loopholes and reinforces the market's role in the overexploitation of sharks (Drakopulos, 2020, p. 106). Furthermore, commercial fishing is another threat to scalloped hammerhead sharks with around 370 tonnes of their species legally allowed to be caught per year in the country (Australian Marine Conservation Society, 2023, p. 2).

2.3. Inadequate Legislative Protection of Scalloped Hammerhead Sharks

According to the Australian Marine Conservation Society (2023, p. 3), several species of hammerhead sharks, not only scalloped hammerhead sharks are inadequately protected from illegal, unreported and unregulated (IUU) activity under current Australian Biodiversity regulations and policy. For example, scalloped hammerhead sharks are still classified under the 'conservation dependent' category under the EPBC Act (Department of Climate Change, Energy, the Environment and Water, 2024b, p. 2). This is despite their low numbers, as already sufficiently proved during the first section of this analysis. The Australian Marine Conservation Society (2023, p. 3) describes ‘conservation dependent’ as an ‘odd category’ which was established to continue the commercial fishing of endangered and threatened aquatic species. Furthermore, under this category, 78 tonnes of scalloped hammerhead sharks can be commercially fished in the Great Barrier Reef which is higher than any other part of the country (Australian Marine Conservation Society, 2023, p. 4). Furthermore, sharks in general are predominantly captured by commercial net fishers, especially in offshore waters due to the lack of regulatory policies in place (Macbeth et al., 2009, p. 4).

3. Analysis of the Environmental Policies

Table 1: An Overview of Policies Targeting the Protection of Scalloped Hammerhead Sharks in Australia

Policy/Regulation	Definition	Advantages	Disadvantages	Conceptual Frameworks/Underpinnings	Effectiveness in Addressing the Problem
Convention of International Trade in Endangered Species of Wild Fauna and Flora (CITES)	Regulates international trade of listed species (in appendices I, II or III), including scalloped hammerheads - through a permitting system.	• Promotes sustainable trade practices. • Encourages international cooperation for conservation efforts.	• Compliance to regulation varies by country and illegal trade persists. • Limited impact on local fisheries and bycatch.	Emphasises market-based solutions; relies on international agreements for compliance.	Moderate effectiveness; has raised awareness but illegal overfishing and bycatch in fisheries continues.
Convention on the Conservation of Migratory Species of Wild Animals (CMS)	Lists migratory species (scalloped hammerhead sharks included) under international agreements to which Australia is a party of, and the listed species are protected under the EPBC Act.	• Provides legal protections for migratory species, including the scalloped hammerhead shark.	• Variability in implementation across different countries.	Emphasises international cooperation between countries with migratory species through legal frameworks.	Effectiveness depends on member state's commitment to implementation and enforcement.
The Environment Protection and Biodiversity Conservation (EPBC) Act 1999	Australia's national environmental legislation which provides a framework for protecting and managing nationally and internationally recognised (threatened or endangered) animals, plants and habitats.	• Legally binding protections for threatened species. • Comprehensive assessment processes for activities affecting biodiversity.	• In 2024, scalloped hammerhead sharks were retained in the 'conservation dependent' category enabling commercial fishing. • Ongoing debates about the adequacy of the 'CD' category.	Neoliberal influences on 'resource management' and 'resource allocation'.	Political inertia and competing economic interests delay necessary actions; the 'conservation dependent' status for scalloped hammerhead shark allows continued overfishing without recovery plans.
The Action Plan for Sharks and Rays 2021	An evaluation framework of the extinction risks for 328 species native to Australian waters; utilises the IUCN Red List Categories at the national level.	• Listed as endangered under the Plan, unlike the EPBC Act.	• Reduction in extinction risk may not mean that the species are entirely out of risk.	Based on scientific assessments and risk evaluation frameworks; aligns well with IUCN guidelines.	Effective; highlighted where Australian efforts were lacking in marine conservation and points where well-funded research has improved conservation outcomes.
The Fisheries (Hammerhead Sharks) Amendment Declaration 2022	Enforces specific legislation and regulation on fishing practices specific to scalloped hammerhead sharks, including requiring heads and fins to remain attached on landing, incorporating the 'fins naturally attached' (FNA) policy which has been effective in EU.	• Directly targets illegal, unreported and unregulated (IUU) fishing practices which affect hammerhead sharks, especially scalloped hammerhead sharks. • Enhances monitoring of shark catches and supports	• Some enforcement challenges still remain, especially regarding bycatch in fisheries. • Operational only in the state of Queensland. • Does allow commercial fishers in QLD to continue	Reflects strong neoliberal principles of resource management; trying to balance economic interests with conservation needs.	Effective in regulating IUU fishing to some degree in QLD, but too soon to access long-term success. Also, broader ecological concerns persist, particularly regarding bycatch and habitat protection.

		sustainable management through better identification of species landed and reducing the risk of shark fining at sea.	retaining hammerhead sharks for economic viability.		
Fins Naturally Attached (FNA) Policy	Requires shark fins to remain attached to their bodies when landed to effectively mitigate live shark fining practices.	• Prevention of live shark fining practices. • Supports more efficient monitoring of catches.	• Although it changes how sharks are brought to land, it still may not reduce killings.	Grounded in principles of sustainability and aligns with broader conservation goals related to marine biodiversity.	High effective globally, backed by the Australian Marine Conservation Society.

Sources: (Australian Marine Conservation Society, 2023; DCCEEW, 2024a; DCCEEW, 2024b; DCCEW, n.d.; Department of Agriculture and Fisheries, 2022)

3.1. The Convention of International Trade in Endangered Species of Wild Fauna and Flora (CITES) and Convention on the Conservation of Migratory Species of Wild Animals (CMS)

The protection of scalloped hammerhead sharks in Australia is shaped by a complex network of Commonwealth, state and international policies. This is further exacerbated by socio-economic pressures and jurisdictional complexities. While international treaties like CITES and CMS provide foundational guidelines for their conservation, implementation is hindered on a local level due to inconsistent compliance with international regulations and insufficient data on migratory patterns. Under CITES, scalloped hammerhead sharks are listed in Appendix II which means that international trade of their meat or product is regulated in order to prevent population decline. Furthermore, this necessitates that any trade involving scalloped hammerheads must be accompanied by a Non-Detriment Finding (NDF) which ensures that export of the species won't affect their long-term survival in the wild (Morton et al., 2022, p. 999). However, despite international recognition of their importance to marine ecosystems and their dwindling numbers, compliance with CITES guidelines remain inconsistent. This is particularly the case in regions where shark fishing contributes to local economies, which further compromises the effectiveness of CITES regulations. As Heid and Márquez-Ramos (2023, p. 2) point out in their research, the effectiveness of CITES in preventing the decline of endangered wildlife is unclear because of how difficult it is to enforce. This is partly due to the vast number of species the regulation covers, but also because of the detailed regulations it requires member nations to implement which end up – increasing costs associated with wildlife trade or making it illegal (Heid & Márquez-Ramos, 2023, p. 2). However, the problem arises when imperfect enforcement mechanisms lead to a shift in wildlife trade from 'legal' to 'illegal' sources (Heid & Márquez-Ramos, 2023, p. 2). Similarly, in Australia, one of the major gaps in policy are the concerns about compliance and reporting in fisheries, with the Department of Climate Change, Energy, the Environment and Water (2024a, p. 17) noting that illegal, unreported and unregulated (IUU) fishing activity was common in

Northern Australia. CMS, on the other hand, focuses on migratory species, including scalloped hammerhead sharks, and emphasises the importance of cross-border conservation. However, lack of data on migratory patterns of scalloped hammerhead sharks limits effective management across jurisdictions under CMS guidelines. This further creates policy gaps in both Australia and neighbouring regions such as Papua New Guinea and Indonesia.

3.2. The Environment Protection and Biodiversity Conservation Act 1999

According to the DCCEEW's 2024 report on 'Listing Advice for Sphyrna lewini (Scalloped Hammerhead), the species was retained in the 'conservation dependent' category under the EPBC Act due to the following reason:

"The main factor that makes the species eligible for listing in the Endangered category is an inferred population reduction of greater than 50% over the last 3 generations (72 years). The Committee's inference is based on fishery-dependent catch information; a suite of candidate stock assessment models; standardised catch rates from the Queensland (Qld) Shark Control Program; and fishery-independent shark surveys in the Java Sea. Fish species may be listed under subsection 179(6) of the EPBC Act as Conservation Dependent if it is the 'focus of a plan of management that provides for management actions necessary to stop the decline of, and support the recovery of, the species so its chances of long-term survival in nature are maximised'. The management arrangements for the scalloped hammerhead are outlined in the National Scalloped Hammerhead Shark Management Strategy 2022 and subsequent Northern Territory and Queensland fisheries legislative amendments."

This reasoning provides several flaws, including the fact that the justification relies heavily on the National Scalloped Hammerhead Shark Management Strategy 2022 which the report claims is designed to health declining populations. However, there is currently no strong evidence that the management strategy will be effective in reversing the population decline. In fact, scalloped hammerhead sharks are still declared 'critically endangered' under the IUCN (IUCN, 2024, p. 2). Furthermore, according to the NSW Government's (2024, p. 1) page on scalloped hammerhead sharks, ‘commercial, recreational and shark meshing bather protection fisheries’ are still one of the primary threats to the species. However, as long as they are classified under ‘conservation dependent’ legal protections and recovery efforts will fail due to commercial fishing being allowed in states like Western Australia (Western Australian Government, 2023, p. 1). According to Lyster et al. (2022, p. 2), the reluctance of Australian governments to protect the environment can be traced back to ideologies of ‘economic rationalism’ which has since evolved into a key element of neoliberalism. Similarly, the history of the EPBC Act can be tracked back to the 1999 Howard Government (Lyster et al., 2022, p. 2). According to Lyster et al. (2022, p. 2), he replaced the previous Whitlam Government’s legislation – which mandated that all government ministers were to consider the environmental impact of any decisions they take under any Act – with the EPBC Act. In stark contrast to the previous Whitlam legislation, the EPBC Act, which came into effect on July 16, 2000, significantly narrowed the scope of federal involvement in environmental issues by focusing only on developments affecting ‘Matters of National Environmental Significance’, i.e., those areas where Australia had international obligations (Lyster et al., 2022, p. 2). These shifting priorities can be observed in Australia’s current environmental governance and how it is shaped by decades of economic rationalism and market-focused policies, which have directly influenced the listing and management of vulnerable species like scalloped hammerhead sharks. Under the EPBA Act, the federal government intervenes selectively, shifting the responsibility to state-led implementation strategies. Furthermore, under the EPBC Act, all three endangered hammerhead species – scalloped hammerhead sharks, great hammerhead sharks, and smooth hammerhead sharks – were evaluated by the Threatened Species Scientific Committee (TSSC) to possibly be listed under the ‘threatened’ category. The decision to retain the scalloped hammerhead in the ‘Conservation Dependent’ category, even after a 2024 review, reflects an ongoing trade-off between conservation efforts and economic interests.

3.3. The Action Plan for Sharks and Rays 2021

According to Kyne et al. (2021, p. 3), the *Action Plan for Sharks and Rays 2021* provides a nationwide assessment of extinction risks for 328 species of chondrichthyans found in Australian marine ecosystems, i.e., sharks, rays, and chimaeras. The Action Plan utilises the 'IUCN Red List of Threatened Species Categories and Criteria' to perform extinction risk assessments for each species on a national scale, which is then assessed based on 'all information available on the species' – including taxonomy, distribution, population status, habitat and ecology, major threats, utilization and trade, and existing conservation measures (Kyne et al., 2021, p. 3). Furthermore, it borrows from the IUCN framework and employs specific thresholds to determine extinction risk, such as (i) population size reduction, (ii) geographic range, and (iii) general population size and (iv) the probability of extinction (Kyne et al., 2021, p. 3). The risk assessments also have diverse range of data sources, including global IUCN evaluations, primary literature, fisheries reports, and stakeholder interviews. The Action Plan was successful in categorising the extinction risk statuses of several species, especially for lesser-known endemics such as whitefin swellsharks, eastern anglesharks and longnose skates, which are species found only in Australian marine ecosystems (Australian Marine Conservation Society, 2021, p. 1).

3.4. The Fisheries (Hammerhead Sharks) Amendment Declaration 2022

The Fisheries (Hammerhead Sharks) Amendment Declaration 2022 is state-implemented management and conservation policy for scalloped hammerhead sharks in Queensland. This amendment is part of broader efforts to address the declining populations of the species. The Declaration imposes stricter restrictions on commercial fishing of hammerhead sharks by prohibiting the sale of the species for commercial fisheries in the state (Queensland Government, 2024, p. 1). Furthermore, while the status of scalloped hammerhead shark remains classified as 'conservation dependent' under the EPBC Act, the 2022 Amendment classifies them as a 'no-take' species (Queensland Government, 2024, p. 1). This essentially means that they are protected species under the Fisheries Act 1994, prohibiting commercial fishing and authorisation would be required to take, keep, use and release of the sharks (Queensland Government, 2024, p. 1). Furthermore, the amendment addresses the ongoing concerns about the species' declining populations, especially in Queensland, where around 2,491 scalloped hammerhead sharks were dumped overboard by commercial fishers in a single year (Australian Marine Conservation Society, 2019b, p. 1).

3.5. Fins Naturally Attached (FNA) Policy

According to the Australian Marine Conservation Society (2019, p. 3), the FNA policy is globally considered to be an effective way to reduce illegal shark finning and preventing endangered sharks, such as scalloped hammerhead sharks, from being traded. Furthermore, FNA ensures that sharks are brought to land without their fins cut off and harvested, which means that no sharks are 'cruelly live finned.' This also guarantees easier identification of shark species as they are extremely hard to identify from just their separated fins, leading to improved data collection (Australian Marine Conservation Society, 2019, p. 3). The FNA policy has also been expanded in Queensland from the East Coast to the Gulf of Carpentaria (Queensland Parliament, 2022, p.4).

4. Recommendations for Policy Change

4.1. Reassessment of the 'Conservation Dependent' Category of Scalloped Hammerhead Sharks under the EPBC Act 1999

This policy recommendation suggests that the current classification of scalloped hammerhead sharks as 'conservation dependent' under the Environment Protection and Biodiversity Conservation (EPBC) Act 1999 be reassessed. According to the Australian Marine Conservation Society (2019, p.1), the current category assigned to scalloped hammerhead sharks allows continued commercial fishing across the country, especially in states like Western Australia where state policy hasn't been formulated yet to protect the species. This severely undermines recovery efforts for the 'critically endangered' species, with an average of 44 tonnes of hammerhead sharks reported to be captured in Western Australia alone from 2015 – 2021 (Department of Climate Change, Energy, the Environment and Water, 2024c, p. 17). However, according to the Australian Marine Conservation Society (2023, p. 1) the reported figures are still significantly lower and underreported. Across the country, other than Queensland and NSW, stringent protocols for protecting the species are missing – especially in Western Australia, where the Ningaloo Reef is a critical habitat for these species (Australian Marine Conservation Society, 2019c, p. 1). A reassessment could lead to a more protective status for the species that prioritises conservation over economic interests, reflecting the urgent need for stronger measures to ensure the survival of scalloped hammerheads in Australian waters. Furthermore, the Threatened Species Scientific Committee (TSSC) and the Threatened Ecological Communities Scientific Committee (TECSC) are key advisory bodies focused on the conservation of species and ecological communities (Department of Climate Change, Energy, the Environment and Water, n.d.-b, p. 1). This policy recommend suggests that the TSSC re-evaluates and provides recommendations on the status of scalloped hammerhead sharks as 'conservation dependent' and instead classify them under 'threatened' species, while the TECSC does the same for ecological communities. Furthermore, this recommendation suggests that the TSSC and the TECSC should review the categorization of scalloped hammerhead sharks and make recommendations to the Minister for Environment at their next annual meeting to amend the EPBC Act.

4.2. List Scalloped Hammerhead Sharks as a 'No-Take' Species

Under the EPBC Act, the scalloped hammerhead shark is still classified as 'conservation dependent' rather than a 'threatened' species, despite a 2024 review (see; Samuel, 2020). This means that they can still be legally and recreationally fished in the country's marine ecosystems. However, both NSW and Queensland have introduced state legislations to declare the scalloped hammerhead shark as an 'endangered' and 'no-take' species, respectively (NSW Government, 2024, p. 1; Queensland Government, 2024, p. 1). For Queensland, the status was given through the amendment of the Queensland Fisheries Act 1994 (Department of Climate Change, Energy, the Environment and Water, 2024c, p. 16). This policy recommendation suggests that following the Australian Martine Conservation Society's recommendations and precedent of the Queensland Government, Scalloped Hammerhead Sharks should be considered as a 'no take' species nationally. Declaring scalloped hammerhead sharks as a 'no-take' species nationally would provide stronger legal protections against overfishing across the country. It will also allow populations to recover more effectively. Furthermore, this aligns with the findings and goals of the Australian Marine Conservation Society (2023, p. 1) who are one of the most vocal advocates for stricter measures to safeguard scalloped hammerhead sharks. By prohibiting all forms of fishing in regard to scalloped hammerhead sharks, including recreational and commercial activities, the policy would help stabilise and potentially recover their numbers over time in the Australian waters.

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